Practical No. 04

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Roll No.:

Title - Write a program for implementing security measures in an IIoT system.

CODE –

```
#include <LiquidCrystal.h>
// Initialize the LCD with the pin numbers
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
int V_GasSen = 0;
int V_TempSens = 0;
void setup() {  pinMode(A0, INPUT); // Gas sensor pin
  pinMode(A1, INPUT); // Temperature sensor pin
  pinMode(7, OUTPUT); // Buzzer pin
  pinMode(9, OUTPUT); // LED for gas detection
  pinMode(12, OUTPUT); // LED for temperature warning
 lcd.begin(16, 2); // Initialize the LCD with 16 columns and 2 rows }
void loop() { // Read gas sensor value
  V_GasSen = analogRead(A0);
// Read temperature sensor value and calculate temperature
  V_TempSens = -40 + 0.488155 * (analogRead(A1) - 20);
// Display temperature and gas status on the LCD
  lcd.clear(); // Clear the LCD
  lcd.setCursor(0, 0); // Set cursor to the first row
  lcd.print("Temperature: "); // Print temperature label
  lcd.print(V_TempSens); // Print temperature value
  lcd.print(" C"); // Print temperature unit
 lcd.setCursor(0, 1); // Set cursor to the second row
  lcd.print("Gas: "); // Print gas label
```

```
  lcd.print(V_GasSen); // Print gas sensor value
 // Check for alerts
  if (V_GasSen >= 250) {  tone(7, 523, 1000); // Play tone if gas is detected
    digitalWrite(9, HIGH); // Turn on the gas detection LED
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ALERT: Gas Detected");
  } else {
    digitalWrite(9, LOW); // Turn off the gas detection LED }
  if (V_TempSens >= 70) {  tone(7, 523, 1000); // Play tone if temperature exceeds the
threshold
    digitalWrite(12, HIGH); // Turn on the temperature warning LED
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ALERT: Temp High");
  } else {
    digitalWrite(12, LOW); // Turn off the temperature warning LED   }
delay(1000); // Delay for one second
}
```

OUTPUT –

